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Roll No

EC-7004(3)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Image Processing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain sampling and quantization. Explain the effects of reducing sampling and quantization.
b) Explain how an image is formed in the human eye and how it adapts and discriminate brightness level?
2. a) Explain Histogram equalization, perform histogram equalization for a 3-bit image ($L=8$) of size 64×64 pixels ($MN = 4096$) has the intensity distribution shown in following table.

r_k	0	1	2	3	4	5	6	7
n_k	123	78	281	417	639	1054	816	688

- b) Construct Arithmetic coding for the string "ABCD" using following table:

Symbols	A	B	C	D
Probability	0.4	0.3	0.2	0.1

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3. a) Explain Huffman coding with the help of word 'CONSTITUTION OF INDIA' find out L_{avg} , Redundancy (R_D), Compression Ratio (C_R), Entropy (H), and efficiency (η).
b) Explain LZW compression technique with example.
4. a) Define of KL transform. Justify that KLT is an optimal transform.
b) Name the categories of Image Enhancement and explain.
5. a) Explain the Geometric Transformations used in image restoration.
b) Explain Weiner filtering approach for image restoration.
6. a) Consider the following binary sequence
000101110010100101.....
Use the Lempel-Ziv coding algorithm to encode this sequence. Assume that the binary symbols 0 and 1 are already in the codebook.
b) Discuss JPEG compression standard.
7. a) Explain Image Restoration filters in detail.
b) Explain 2-D discrete Fourier Transform of image processing.
8. Write short notes:
 - a) Digital Image Watermarking
 - b) Golomb coding
 - c) Walsh Transform
 - d) Image sampling and Quantization

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